

No. 2626/QĐ-BTNMT

Hanoi, October 10, 2022

DECISION

Announcing the list of emission factors for greenhouse gas inventory

THE MINISTER OF NATURAL RESOURCES AND ENVIRONMENT

Pursuant to the Law on Environment Protection dated November 17, 2020;

Pursuant to the Government's Decree No. 36/2017/ND-CP dated April 4, 2017, defining the functions, tasks, powers and organizational structure of the Ministry of Natural Resources and Environment;

Pursuant to Decree No. 06/2022/ND-CP dated January 7, 2022, of the Government providing regulations on mitigation of greenhouse gas emissions and protection of the ozone layer;

At the proposal of the Director General of the Department of Climate Change,

DECIDES:

Article 1. To announce the list of emission factors for greenhouse gas inventory, including:

1. List of emission factors for greenhouse gas inventory - Energy in Appendix I;
2. List of emission factors for greenhouse gas inventory - Industrial processes and product use in Appendix II;
3. List of emission factors for greenhouse gas inventory - Agriculture, forestry and other land use in Appendix III;
4. List of emission factors for greenhouse gas inventory - Waste in Appendix IV.

Article 2. To assign the Department of Climate Change to assume the prime responsibility for reviewing and updating the list of emission factors when there is a new country-specific greenhouse gas emission factor, or when there is an updated greenhouse gas inventory guideline from the Intergovernmental Panel

on Climate Change, submit to the Minister of Natural Resources and Environment for decision on updating the list.

Article 3. This Decision takes effect on the date of its signing.

The Chief of Office, the Director General of Climate Change and relevant agencies and organizations shall implement this Decision./.

**FOR THE MINISTER
THE DEPUTY MINISTER**

Le Cong Thanh

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APPENDIX I

LIST OF EMISSION FACTORS FOR GREENHOUSE GAS INVENTORY - ENERGY

*(Attached to Decision No. 2626/QĐ-BTNMT dated October 10, 2022, of the
Minister of Natural Resources and Environment)*

No.	Name of greenhouse gas emission factor	Type of greenhouse gas emission	Emission source	Value	Unit	Method applied under IPCC Guidelines
1.	Fuel combustion activities					
1.1	CO ₂ emission factor of anthracite coal	CO ₂	Energy industry	98,300	Kg CO ₂ /TJ	Tier 1
1.2	CH ₄ emission factor of anthracite coal	CH ₄	Energy industry	1	Kg CH ₄ /TJ	Tier 1
1.3	N ₂ O emission factor of anthracite coal	N ₂ O	Energy industry	1.5	Kg N ₂ O/TJ	Tier 1
1.4	CO ₂ emission factor of sub-bituminous coal	CO ₂	Energy industry	96,100	Kg CO ₂ /TJ	Tier 1
1.5	CH ₄ emission factor of sub-	CH ₄	Energy industry	1	Kg CH ₄ /TJ	Tier 1

	bituminous coal					
1.6	N ₂ O emission factor of sub-bituminous coal	N ₂ O	Energy industry	1.5	Kg N ₂ O/TJ	Tier 1
1.7	CO ₂ emission factor of crude oil	CO ₂	Energy industry	73,300	Kg CO ₂ /TJ	Tier 1
1.8	CH ₄ emission factor of crude oil	CH ₄	Energy industry	3	Kg CH ₄ /TJ	Tier 1
1.9	N ₂ O emission factor of crude oil	N ₂ O	Energy industry	0.6	Kg N ₂ O/TJ	Tier 1
1.10	CO ₂ emission factor of diesel oil	CO ₂	Energy industry	74,100	Kg CO ₂ /TJ	Tier 1
1.11	CH ₄ emission factor of diesel oil	CH ₄	Energy industry	3	Kg CH ₄ /TJ	Tier 1
1.12	N ₂ O emission factor of diesel oil	N ₂ O	Energy industry	0.6	Kg N ₂ O/TJ	Tier 1
1.13	CO ₂ emission factor of fuel oil	CO ₂	Energy industry	77,400	Kg CO ₂ /TJ	Tier 1
1.14	CH ₄ emission	CH ₄	Energy industry	3	Kg CH ₄ /TJ	Tier 1

	factor of fuel oil					
1.15	N ₂ O emission factor of fuel oil	N ₂ O	Energy industry	0.6	Kg N ₂ O/TJ	Tier 1
1.16	CO ₂ emission factor of natural gas	CO ₂	Energy industry	56,100	Kg CO ₂ /TJ	Tier 1
1.17	CH ₄ emission factor of natural gas	CH ₄	Energy industry	1	Kg CH ₄ /TJ	Tier 1
1.18	N ₂ O emission factor of natural gas	N ₂ O	Energy industry	0.1	Kg N ₂ O/TJ	Tier 1
1.19	CO ₂ emission factor of biomass	CO ₂	Energy industry	100,000	Kg CO ₂ /TJ	Tier 1
1.20	CH ₄ emission factor of biomass	CH ₄	Energy industry	30	Kg CH ₄ /TJ	Tier 1
1.21	N ₂ O emission factor of biomass	N ₂ O	Energy industry	4	Kg N ₂ O/TJ	Tier 1
1.22	CO ₂ emission factor of charcoal	CO ₂	Energy industry	112,000	Kg CO ₂ /TJ	Tier 1
1.23	CH ₄ emission factor of charcoal	CH ₄	Energy industry	200	Kg CH ₄ /TJ	Tier 1

1.24	N ₂ O emission factor of charcoal	N ₂ O	Energy industry	4	Kg N ₂ O/TJ	Tier 1
1.25	CO ₂ emission factor of anthracite coal	CO ₂	Manufacturin g and construction industry	98,300	Kg CO ₂ /TJ	Tier 1
1.26	CH ₄ emission factor of anthracite coal	CH ₄	Manufacturin g and construction industry	10	Kg CH ₄ /TJ	Tier 1
1.27	N ₂ O emission factor of anthracite coal	N ₂ O	Manufacturin g and construction industry	1.5	Kg N ₂ O/TJ	Tier 1
1.28	CO ₂ emission factor of diesel oil	CO ₂	Manufacturin g and construction industry	74,100	Kg CO ₂ /TJ	Tier 1
1.29	CH ₄ emission factor of diesel oil	CH ₄	Manufacturin g and construction industry	3	Kg CH ₄ /TJ	Tier 1
1.30	N ₂ O emission factor of diesel oil	N ₂ O	Manufacturin g and construction industry	0.6	Kg N ₂ O/TJ	Tier 1
1.31	CO ₂ emission factor of fuel oil	CO ₂	Manufacturin g and construction industry	77,400	Kg CO ₂ /TJ	Tier 1
1.32	CH ₄ emission factor of fuel oil	CH ₄	Manufacturin g and construction industry	3	Kg CH ₄ /TJ	Tier 1

1.33	N ₂ O emission factor of fuel oil	N ₂ O	Manufacturin g and construction industry	0.6	Kg N ₂ O/TJ	Tier 1
1.34	CO ₂ emission factor of liquefied petroleum gas	CO ₂	Manufacturin g and construction industry	63,100	Kg CO ₂ /TJ	Tier 1
1.35	CH ₄ emission factor of liquefied petroleum gas	CH ₄	Manufacturin g and construction industry	1	Kg CH ₄ /TJ	Tier 1
1.36	N ₂ O emission factor of liquefied petroleum gas	N ₂ O	Manufacturin g and construction industry	0.1	Kg N ₂ O/TJ	Tier 1
1.37	CO ₂ emission factor of natural gas	CO ₂	Manufacturin g and construction industry	56,100	Kg CO ₂ /TJ	Tier 1
1.38	CH ₄ emission factor of natural gas	CH ₄	Manufacturin g and construction industry	1	Kg CH ₄ /TJ	Tier 1
1.39	N ₂ O emission factor of natural gas	N ₂ O	Manufacturin g and construction industry	0.1	Kg N ₂ O/TJ	Tier 1
1.40	CH ₄ emission factor of biomass	CH ₄	Manufacturin g and construction industry	30	Kg CH ₄ /TJ	Tier 1

1.41	N ₂ O emission factor of biomass	N ₂ O	Manufacturin g and construction industry	4	Kg N ₂ O/TJ	Tier 1
1.42	CO ₂ emission factor of jet kerosene	CO ₂	Domestic air transportation	71,500	Kg CO ₂ /TJ	Tier 1
1.43	CO ₂ emission factor of aviation gasoline	CO ₂	Domestic air transportation	70,000	Kg CO ₂ /TJ	Tier 1
1.44	CH ₄ emission factor of all fuels	CH ₄	Domestic air transportation	0.5	Kg CH ₄ /TJ	Tier 1
1.45	N ₂ O emission factor of all fuels	N ₂ O	Domestic aviation	2	Kg N ₂ O/TJ	Tier 1
1.46	CO ₂ emission factor of diesel oil	CO ₂	Road transportation	74,100	Kg CO ₂ /TJ	Tier 1
1.47	CH ₄ emission factor of diesel oil	CH ₄	Road transportation	3.9	Kg CH ₄ /TJ	Tier 1
1.48	N ₂ O emission factor of diesel oil	N ₂ O	Road transportation	3.9	Kg N ₂ O/TJ	Tier 1
1.49	CO ₂ emission factor of gasoline	CO ₂	Road transportation	69,300	Kg CO ₂ /TJ	Tier 1
1.50	CH ₄ emission	CH ₄	Road transportation	33	Kg CH ₄ /TJ	Tier 1

	factor of gasoline					
1.51	N ₂ O emission factor of gasoline	N ₂ O	Road transportation	3.2	Kg N ₂ O/TJ	Tier 1
1.52	CO ₂ emission factor of liquefied petroleum gas	CO ₂	Road transportation	63,100	Kg CO ₂ /TJ	Tier 1
1.53	CH ₄ emission factor of liquefied petroleum gas	CH ₄	Road transportation	62	Kg CH ₄ /TJ	Tier 1
1.54	N ₂ O emission factor of liquefied petroleum gas	N ₂ O	Road transportation	0.2	Kg N ₂ O/TJ	Tier 1
1.55	CO ₂ emission factor of natural gas	CO ₂	Road transportation	56,100	Kg CO ₂ /TJ	Tier 1
1.56	CH ₄ emission factor of natural gas	CH ₄	Road transportation	92	Kg CH ₄ /TJ	Tier 1
1.57	N ₂ O emission factor of natural gas	N ₂ O	Road transportation	3	Kg N ₂ O/TJ	Tier 1
1.58	CO ₂ emission	CO ₂	Railways	74,100	Kg CO ₂ /TJ	Tier 1

	factor of diesel oil					
1.59	CH ₄ emission factor of diesel oil	CH ₄	Railways	4.15	Kg CH ₄ /TJ	Tier 1
1.60	N ₂ O emission factor of diesel oil	N ₂ O	Railways	28.6	Kg N ₂ O/TJ	Tier 1
1.61	CO ₂ emission factor of diesel oil	CO ₂	Inland waterway and maritime transportation	74,100	Kg CO ₂ /TJ	Tier 1
1.62	CH ₄ emission factor of diesel oil	CH ₄	Inland waterway and maritime transportation	7	Kg CH ₄ /TJ	Tier 1
1.63	N ₂ O emission factor of diesel oil	N ₂ O	Inland waterway and maritime transportation	2	Kg N ₂ O/TJ	Tier 1
1.64	CO ₂ emission factor of fuel oil	CO ₂	Inland waterway and maritime transportation	77,400	Kg CO ₂ /TJ	Tier 1
1.65	CH ₄ emission factor of fuel oil	CH ₄	Inland waterway and maritime transportation	7	Kg CH ₄ /TJ	Tier 1
1.66	N ₂ O emission factor of fuel oil	N ₂ O	Inland waterway and maritime transportation	2	Kg N ₂ O/TJ	Tier 1
1.67	CO ₂ emission factor of anthracite coal	CO ₂	Trade and services	98,300	Kg CO ₂ /TJ	Tier 1

1.68	CH ₄ emission factor of anthracite coal	CH ₄	Trade and services	10	Kg CH ₄ /TJ	Tier 1
1.69	N ₂ O emission factor of anthracite coal	N ₂ O	Trade and services	1.5	Kg N ₂ O/TJ	Tier 1
1.70	CO ₂ emission factor of diesel oil	CO ₂	Trade and services	74,100	Kg CO ₂ /TJ	Tier 1
1.71	CH ₄ emission factor of diesel oil	CH ₄	Trade and services	10	Kg CH ₄ /TJ	Tier 1
1.72	N ₂ O emission factor of diesel oil	N ₂ O	Trade and services	0.6	Kg N ₂ O/TJ	Tier 1
1.73	CO ₂ emission factor of liquefied petroleum gas	CO ₂	Trade and services	63,100	Kg CO ₂ /TJ	Tier 1
1.74	CH ₄ emission factor of liquefied petroleum gas	CH ₄	Trade and services	5	Kg CH ₄ /TJ	Tier 1
1.75	N ₂ O emission factor of liquefied	N ₂ O	Trade and services	0.1	Kg N ₂ O/TJ	Tier 1

	petroleum gas					
1.76	CO ₂ emission factor of charcoal	CO ₂	Trade and services	112,000	Kg CO ₂ /TJ	Tier 1
1.77	CH ₄ emission factor of charcoal	CH ₄	Trade and services	200	Kg CH ₄ /TJ	Tier 1
1.78	N ₂ O emission factor of charcoal	N ₂ O	Trade and services	1	Kg N ₂ O/TJ	Tier 1
1.79	CO ₂ emission factor of anthracite coal	CO ₂	Civil	98,300	Kg CO ₂ /TJ	Tier 1
1.80	CH ₄ emission factor of anthracite coal	CH ₄	Civil	300	Kg CH ₄ /TJ	Tier 1
1.81	N ₂ O emission factor of anthracite coal	N ₂ O	Civil	1.5	Kg N ₂ O/TJ	Tier 1
1.82	CO ₂ emission factor of kerosene	CO ₂	Civil	71,900	Kg CO ₂ /TJ	Tier 1
1.83	CH ₄ emission factor of kerosene	CH ₄	Civil	10	Kg CH ₄ /TJ	Tier 1
1.84	N ₂ O emission	N ₂ O	Civil	0.6	Kg N ₂ O/TJ	Tier 1

	factor of kerosene					
1.85	CO ₂ emission factor of liquefied petroleum gas	CO ₂	Civil	63,100	Kg CO ₂ /TJ	Tier 1
1.86	CH ₄ emission factor of liquefied petroleum gas	CH ₄	Civil	5	Kg CH ₄ /TJ	Tier 1
1.87	N ₂ O emission factor of liquefied petroleum gas	N ₂ O	Civil	0.1	Kg N ₂ O/TJ	Tier 1
1.88	CO ₂ emission factor of biomass	CO ₂	Civil	100,000	Kg CO ₂ /TJ	Tier 1
1.89	CH ₄ emission factor of biomass	CH ₄	Civil	300	Kg CH ₄ /TJ	Tier 1
1.90	N ₂ O emission factor of biomass	N ₂ O	Civil	4	Kg N ₂ O/TJ	Tier 1
1.91	CO ₂ emission factor of charcoal	CO ₂	Civil	112,000	Kg CO ₂ /TJ	Tier 1
1.92	CH ₄ emission	CH ₄	Civil	200	Kg CH ₄ /TJ	Tier 1

	factor of charcoal					
1.93	N ₂ O emission factor of charcoal	N ₂ O	Civil	1	Kg N ₂ O/TJ	Tier 1
1.94	CO ₂ emission factor of gasoline	CO ₂	Agriculture, forestry and fisheries	69,300	Kg CO ₂ /TJ	Tier 1
1.95	CH ₄ emission factor of gasoline	CH ₄	Agriculture, forestry and fisheries	10	Kg CH ₄ /TJ	Tier 1
1.96	N ₂ O emission factor of gasoline	N ₂ O	Agriculture, forestry and fisheries	0.6	Kg N ₂ O/TJ	Tier 1
1.97	CO ₂ emission factor of diesel oil	CO ₂	Agriculture, forestry and fisheries	74,100	Kg CO ₂ /TJ	Tier 1
1.98	CH ₄ emission factor of diesel oil	CH ₄	Agriculture, forestry and fisheries	10	Kg CH ₄ /TJ	Tier 1
1.99	N ₂ O emission factor of diesel oil	N ₂ O	Agriculture, forestry and fisheries	0.6	Kg N ₂ O/TJ	Tier 1
1,100	CH ₄ emission factor of biomass	CH ₄	Agriculture, forestry and fisheries	300	Kg CH ₄ /TJ	Tier 1
1,101	N ₂ O emission factor of biomass	N ₂ O	Agriculture, forestry and fisheries	4	Kg N ₂ O/TJ	Tier 1

2 Emissions due to dispersion						
2.1	CH ₄ dispersion factor in underground coal mining*	CH ₄	Underground coal mining	1.5789	m ³ CH ₄ /ton	Tier 2
2.2	CH ₄ dispersion factor after underground coal mining*	CH ₄	Underground coal mining	0.1697	m ³ CH ₄ /ton	Tier 2
2.3	CH ₄ dispersion factor in open-pit coal mining*	CH ₄	Open-pit coal mining	0.05375	m ³ CH ₄ /ton	Tier 2
2.4	CH ₄ dispersion factor after open-pit coal mining*	CH ₄	Open-pit coal mining	0.1697	m ³ CH ₄ /ton	Tier 2
2.5	CO ₂ emission factor leaked from oil production	CO ₂	Oil extraction	0.00215	Thousand tons of CO ₂ /10 ³ m ³ of total oil products	Tier 1
2.6	CH ₄ emission factor leaked from oil production	CH ₄	Oil extraction	0.01035	Thousand tons of CH ₄ /10 ³ m ³ of total oil products	Tier 1
2.7	CO ₂ emission factor due to	CO ₂	Oil extraction	0.0405	Thousand tons of CO ₂ /10 ³ m ³	Tier 1

	natural combustion from oil production				of total oil products	
2.8	CH ₄ emission factor due to natural combustion from oil production	CH ₄	Oil extraction	0.000025	Thousand tons of CH ₄ /10 ³ m ³ of total oil products	Tier 1
2.9	N ₂ O emission factor due to natural combustion in oil production	N ₂ O	Oil extraction	0.00000064	Thousand tons/10 ³ m ³ of total oil products	Tier 1
2.10	CO ₂ emission factor released in oil production	CO ₂	Oil extraction	0.00249	Thousand tons of CO ₂ /10 ³ m ³ of total oil products	Tier 1
2.11	CH ₄ emission factor released in oil production	CH ₄	Oil extraction	0.0196	Thousand tons of CH ₄ /10 ³ m ³ of total oil products	Tier 1
2.12	CO ₂ emission factor leaked in gas treatment	CO ₂	Natural gas extraction	0.0675	Thousand tons of CO ₂ /10 ⁶ m ³ of total input raw gas	Tier 1
2.13	CO ₂ emission factor due to natural	CO ₂	Natural gas extraction	0.00355	Thousand tons of CO ₂ /10 ⁶ m ³	Tier 1

	combustion in gas treatment				of total gas products	
2.14	CH ₄ emission factor due to natural combustion in gas treatment	CH ₄	Natural gas extraction	0.0000024	Thousand tons of CH ₄ /10 ⁶ m ³ of total gas products	Tier 1
2.15	CH ₄ emission factor due to natural combustion in gas treatment	N ₂ O	Natural gas extraction	3,9E-08	Thousand tons of N ₂ O/10 ⁶ m ³ of total gas products	Tier 1
2.16	CO ₂ emission factor due to natural combustion in gas production	CO ₂	Natural gas extraction	0.0014	Thousand tons of CO ₂ /10 ⁶ m ³ of total gas products	Tier 1
2.17	CH ₄ emission factor due to natural combustion in gas production	CH ₄	Natural gas extraction	0.0000008 8	Thousand tons of CH ₄ /10 ⁶ m ³ of total gas products	Tier 1
2.18	N ₂ O emission factor due to natural combustion in gas production	N ₂ O	Natural gas extraction	2,5E-08	Thousand tons of N ₂ O/10 ⁶ m ³ of total gas products	Tier 1
2.19	CO ₂ emission	CO ₂	Natural gas extraction	0.000097	Thousand tons of	Tier 1

	factor released in gas production				CO ₂ /10 ⁶ m ³ of total gas products	
2.20	CH ₄ emission factor released in gas production	CH ₄	Natural gas extraction	0.01219	Thousand tons of CH ₄ /10 ⁶ m ³ of total gas products	Tier 1
2.21	CO ₂ emission factor released in gas treatment	CO ₂	Natural gas extraction	0.00025	Thousand tons of CO ₂ /10 ⁶ m ³ of total input raw gas products	Tier 1
2.22	CH ₄ emission factor released in gas treatment	CH ₄	Natural gas extraction	0.00079	Thousand tons of CH ₄ /10 ⁶ m ³ of total input raw gas products	Tier 1

(*): Country-specific greenhouse gas emission factor.

APPENDIX II

LIST OF EMISSION FACTORS FOR GREENHOUSE GAS INVENTORY - INDUSTRIAL PROCESSES AND PRODUCT USE

(Attached to Decision No. 2626/QĐ-BTNMT dated October 10, 2022, of the
Minister of Natural Resources and Environment)

No.	Name of greenhouse gas emission factor	Type of greenhouse gas emission	Emission source	Value	Unit	Method applied under IPCC Guidelines
1 Mineral industry						
1.1	CO ₂ emission factor of clinker production mix ²	CO ₂	Cement production process	0.525	Tons of CO ₂ /ton ₂ /ton of clinker	Tier 1
1.2	Content of CaO/clinker	CO ₂	Cement production process	65	%	Tier 1
1.3	CKD correction factor	CO ₂	Cement production process	1.02	-	Tier 1
1.4	CO ₂ emission factor of raw materials for producing quicklime with high calcium content (% CaO: 93-98, % MgO: 0.3-2.5)	CO ₂	Lime production process	0.75	Tons of CO ₂ /ton of lime	Tier 1
1.5	CO ₂ emission factor of raw materials for producing dolomite quicklime (% CaO: 55-57; % MgO: 38-41)	CO ₂	Lime production process	0.77	Tons of CO ₂ /ton of lime	Tier 1

1.6	CO ₂ emission factor of glass production process	CO ₂	Glass production process	0.21	Tons of CO ₂ /ton of float glass	Tier 1
2	Chemical industry					
2.1	Coal consumption factor of partial oxidation technology	CO ₂	Ammonia production process	42.5	GJ/ton NH ₃	Tier 1
2.2	Natural gas consumption factor of gas processing technology	CO ₂	Ammonia production process	37.5	GJ/ton NH ₃	Tier 1
2.3	Factor of carbon content in coal of partial oxidation technology	CO ₂	Ammonia production process	21	Kg C/GJ	Tier 1
2.4	Factor of carbon content in natural gas of gas processing technology	CO ₂	Ammonia production process	15.30	Kg C/GJ	Tier 1
2.5	Carbon oxidation rate of partial oxidation technology	CO ₂	Ammonia production process	1	-	Tier 1
2.6	Carbon oxidation rate of gas processing technology	CO ₂	Ammonia production process	1	-	Tier 1
2.7	N ₂ O emission factor for HNO ₃ production process with medium pressure technology	N ₂ O	Acid nitric production process	8	Kg N ₂ O/ton HNO ₃	Tier 1
3	Metallurgy					

3.1	CO ₂ emission factor with BOF blower technology*	CO ₂	Iron and steel production process	2.47	Tons of CO ₂ /ton of crude steel	Tier 2
3.2	CO ₂ emission factor with EAF electric arc furnace technology*	CO ₂	Iron and steel production process	0.06	Tons of CO ₂ /ton of crude steel	Tier 2

(*): Country-specific greenhouse gas emission factor.

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APPENDIX III

LIST OF EMISSION FACTORS FOR GREENHOUSE GAS INVENTORY - AGRICULTURE, FORESTRY AND OTHER LAND USE

(Attached to Decision No. 2626/QĐ-BTNMT dated October 10, 2022, of the
Minister of Natural Resources and Environment)

No.	Name of greenhouse gas emission factor	Type of greenhouse gas emission	Emission source	Value	Unit	Method applied under IPCC Guidelines
1	Breed					
1.1	CH ₄ emission factor of dairy cow	CH ₄	Food digestion	78	Kg CH ₄ /livestock/year	Tier 1
1.2	CH ₄ emission factor of beef cattle	CH ₄	Food digestion	54	Kg CH ₄ /livestock/year	Tier 1
1.3	CH ₄ emission factor of buffalo	CH ₄	Food digestion	76	Kg CH ₄ /livestock/year	Tier 1
1.4	CH ₄ emission factor of sheep	CH ₄	Food digestion	5	Kg CH ₄ /livestock/year	Tier 1
1.5	CH ₄ emission factor of goat	CH ₄	Food digestion	5	Kg CH ₄ /livestock/year	Tier 1
1.6	CH ₄ emission factor of horse	CH ₄	Food digestion	18	Kg CH ₄ /livestock/year	Tier 1

1.7	CH ₄ emission factor of pig	CH ₄	Food digestion	1	Kg CH ₄ /livestock/year	Tier 1
1.8	Excretion rate of volatile solids from dairy cow waste	CH ₄	Livestock manure management	8.1	Kg/1000kg/day	Tier 2
1.9	Maximum ability to produce CH ₄ from dairy cow waste	CH ₄	Livestock manure management	0.13	m ³ CH ₄ /kg of volatile solids (VS)	Tier 2
1.10	Excretion rate of volatile solids from beef cattle waste	CH ₄	Livestock manure management	11.3	Kg/1000kg/day	Tier 2
1.11	Maximum ability to produce CH ₄ from beef cattle waste	CH ₄	Livestock manure management	0.13	m ³ CH ₄ /kg VS	Tier 2
1.12	Excretion rate of volatile solids from buffalo waste	CH ₄	Livestock manure management	13.1	Kg/1000kg/day	Tier 2
1.13	Maximum ability to produce CH ₄ from buffalo waste	CH ₄	Livestock manure management	0.10	m ³ CH ₄ /kg VS	Tier 2
1.14	Excretion rate of volatile	CH ₄	Livestock manure management	8.30	Kg/1000 kg/day	Tier 2

	solids from sheep waste					
1.1 5	Maximum ability to produce CH ₄ from sheep waste	CH ₄	Livestock manure management	0.13	m ³ CH ₄ /kg VS	Tier 2
1.1 6	Excretion rate of volatile solids from goat waste	CH ₄	Livestock manure management	10.40	Kg/1000kg/day	Tier 2
1.1 7	Maximum ability to produce CH ₄ from goat waste	CH ₄	Livestock manure management	0.13	m ³ CH ₄ /kg VS	Tier 2
1.1 8	Excretion rate of volatile solids from horse waste	CH ₄	Livestock manure management	7.20	Kg/1000 kg/day	Tier 2
1.1 9	Maximum ability to produce CH ₄ from horse waste	CH ₄	Livestock manure management	0.26	m ³ CH ₄ /kg VS	Tier 2
1.2 0	Excretion rate of volatile solids from pig waste	CH ₄	Livestock manure management	5.80	Kg/1000 kg/day	Tier 2
1.2 1	Maximum ability to produce CH ₄ from pig waste	CH ₄	Livestock manure management	0.29	m ³ CH ₄ /kg VS	Tier 2
1.2 2	Excretion rate of	CH ₄	Livestock manure management	11.20	Kg/1000 kg/day	Tier 2

	volatile solids from poultry waste					
1.2.3	Maximum ability to produce CH ₄ from poultry waste	CH ₄	Livestock manure management	0.24	m ³ CH ₄ /kg VS	Tier 2
1.2.4	CH ₄ conversion factor of livestock waste treated with a composting system for an average temperature range of 15-25°C	CH ₄	Livestock manure management	1.00	%	Tier 2
1.2.5	CH ₄ conversion factor of livestock waste treated with aerobic treatment systems for an average temperature range of 15-25°C	CH ₄	Livestock manure management	0.0	%	Tier 2
1.2.6	CH ₄ conversion factor of 15-25°C bio-buffer system	CH ₄	Livestock manure management	1.50	%	Tier 2
1.2.7	CH ₄ conversion factor of livestock waste treated	CH ₄	Livestock manure management	12.97	%	Tier 2

	by anaerobic digester system for an average temperature range of 15-25°C					
1.28	CH ₄ conversion of livestock manure treated with 15-25°C pasture and grazing system	CH ₄	Livestock manure management	1.50	%	Tier 2
1.29	CH ₄ conversion factor of livestock waste treated with a composting system for an average temperature above 25°C	CH ₄	Livestock manure management	1.50	%	Tier 2
1.30	CH ₄ conversion factor of livestock waste treated with aerobic treatment systems for an average temperature above 25°C	CH ₄	Livestock manure management	0.0	%	Tier 2
1.31	CH ₄ conversion factor of	CH ₄	Livestock manure management	1.50	%	Tier 2

	livestock waste treated with bio-buffer system for an average temperature above 25°C					
1.3.2	CH ₄ conversion factor of livestock waste treated by anaerobic digester system for an average temperature above 25°C	CH ₄	Livestock manure management	13.17	%	Tier 2
1.3.3	CH ₄ conversion of livestock manure treated with pasture and grazing system for an average temperature above 25°C	CH ₄	Livestock manure management	2.00	%	Tier 2
1.3.4	N excretion rate of dairy cow	N ₂ O	Livestock manure management (N ₂ O direct emission)	0.47	Kg N/1000 kg of weight/day	Tier 2
1.3.5	N excretion rate of beef cattle	N ₂ O	Livestock manure management (N ₂ O direct emission)	0.34	Kg N/1000 kg of weight/day	Tier 2
1.3.6	N excretion rate of buffalo	N ₂ O	Livestock manure management (N ₂ O direct emission)	0.32	Kg N/1000 kg of weight/day	Tier 2

1.37	N excretion rate of sheep	N ₂ O	Livestock manure management (N ₂ O direct emission)	1.17	Kg N/1000 kg of weight/day	Tier 2
1.38	N excretion rate of goat	N ₂ O	Livestock manure management (N ₂ O direct emission)	1.37	Kg N/1000 kg of weight/day	Tier 2
1.39	N excretion rate of horse	N ₂ O	Livestock manure management (N ₂ O direct emission)	0.46	Kg N/1000 kg of weight/day	Tier 2
1.40	N excretion rate of pig	N ₂ O	Livestock manure management (N ₂ O direct emission)	0.42	Kg N/1000 kg of weight/day	Tier 2
1.41	N excretion rate of poultry	N ₂ O	Livestock manure management (N ₂ O direct emission)	1.1	Kg N/1000 kg of weight/day	Tier 2
1.42	Emission factor of livestock manure treated by composting system for fertilizer	N ₂ O	Livestock manure management (N ₂ O direct emission)	0.01	Kg N ₂ O-N/kg N	Tier 2
1.43	Emission factor of livestock manure treated by aerobic treatment system	N ₂ O	Livestock manure management (N ₂ O direct emission)	0.01	Kg N ₂ O-N/kg N	Tier 2
1.44	Emission factor of livestock manure treated by bio-buffer system	N ₂ O	Livestock manure management (N ₂ O direct emission)	0.001	Kg N ₂ O-N/kg N	Tier 2

1.45	Emission factor of livestock manure treated by anaerobic digester system	N ₂ O	Livestock manure management (N ₂ O direct emission)	0	Kg N ₂ O-N/kg N	Tier 2
2 Land						
2.1	Carbon stocks in the above-ground forest biomass	CO ₂	Forest land remaining forest land	0.47	t-C/t-d.m	Tier 2, Increase-Decrease Method
2.2	Biomass conversion and expansion factor (BCEFi) of natural timber forest	CO ₂	Forest land remaining forest land	0.87	t-d.m/m ³	Tier 2, Increase-Decrease Method
2.3	BCEFi of bamboo forest	CO ₂	Forest land remaining forest land	0.93	t-d.m/m ³	Tier 2, Increase-Decrease Method
2.4	BCEFi of mixed forest	CO ₂	Forest land remaining forest land	0.87	t-d.m/m ³	Tier 2, Increase-Decrease Method
2.5	BCEFi of mangrove forest	CO ₂	Forest land remaining forest land	0.87	t-d.m/m ³	Tier 2, Increase-Decrease Method
2.6	CO ₂ emission factor for drained organic soil	CO ₂	Forest land remaining forest land	1.36	tC/ha/year	Tier 1

	of managed forest land					
2.7	Above-ground biomass carbon before conversion of annual cropland	CO ₂	Cropland converted to other land use classifications	4.7	tC/ha	Tier 1
2.8	Above-ground biomass carbon before conversion of perennial cropland	CO ₂	Cropland converted to other land use classifications	21	tC/ha	Tier 1
2.9	Biomass before conversion of grassland	CO ₂	Land converted to forest land/cropland/wetland, residential land, other land	3.65	t-d.m/ha	Tier 1
2.10	Biomass carbon on annual cropland after one year	CO ₂	Land converted to cropland	4.7	tC/ha/year	Tier 1
2.11	CO ₂ emission factor for cultivated organic soil	CO ₂	Cropland remaining cropland/Land converted to cropland	20	tC/ha/year	Tier 1
2.12	Biomass carbon on perennial cropland after one year	CO ₂	Land converted to cropland	2.60	tC/ha/year	Tier 1

2.1	CO ₂ emission factor for drained organic grassland	CO ₂	Grassland remaining grassland/Land converted to grassland	5	tC/ha/year	Tier 1
3 Other emission sources and emissions other than CO₂						
3.1	Biomass burning factor	CH ₄ , N ₂ O	Emissions from biomass burning in forest land/land converted to other land	0.55	-	Tier 1
3.2	CH ₄ emission factor of burnt dry matter	CH ₄	Emissions from biomass burning in forest land/forest land converted to other land	6.80	g/kgdm	Tier 1
3.3	N ₂ O emission factor of burnt dry matter	N ₂ O	Emissions from biomass burning in forest land/forest land converted to other land	0.20	g/kgdm	Tier 1
3.4	CH ₄ emission factor of burnt dry matter	CH ₄	Emissions from burning biomass from agricultural by-products in the field	2.70	g/kgdm	Tier 1
3.5	N ₂ O emission factor of burnt dry matter	N ₂ O	Emissions from burning biomass from agricultural by-products in the field	0.07	g/kgdm	Tier 1
3.6	CH ₄ emission factor of burnt dry matter	CH ₄	Emissions from burning grassland	2.30	g/kgdm	Tier 1
3.7	N ₂ O emission	N ₂ O	Emissions from burning grassland	0.21	g/kgdm	Tier 1

	factor of burnt dry matter					
3.8	CO ₂ emission factor of calcium lime	CO ₂	Lime used in agricultural cultivation	0.12	Ton C/ton of calcium lime	Tier 1
3.9	CO ₂ emission factor of dolomite lime	CO ₂	Lime used in agricultural cultivation	0.13	Ton C/ton of dolomite lime	Tier 1
3.10	CO ₂ emission factor of nitrogen fertilizer use	CO ₂	Urea fertilizer used in agricultural cultivation	0.20	Ton C/ton of urea	Tier 1
3.11	Factor for estimating the amount of N from fertilizers, livestock manure, and agricultural by-products (EF ₁)	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	0.01	Kg N ₂ O-N/ (kg N)	Tier 1
3.12	Factor for estimating the amount of N from grazing for cattle (cow, dairy cow, buffalo), poultry, pig (EF _{3PRP})	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	0.02	Kg N ₂ O-N/ (kg N)	Tier 1
3.13	Factor for estimating the amount of N from	N ₂ O	Emissions from agricultural by-products, livestock manure,	0.01	Kg N ₂ O-N/ (kg N)	Tier 1

	grazing for sheep and other animals (EF _{3PRP})		pasture/grazing wastes			
3.1 4	N ₂ O-N emission factor from spring/winter-spring rice in the North*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	0.34	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.1 5	N ₂ O-N emission factor from summer-autumn/rice crop in the North*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	0.25	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.1 6	N ₂ O-N emission factor from spring/winter-spring rice in the Central region*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	0.24	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.1 7	N ₂ O-N emission factor from summer-autumn/rice crop in the Central region*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	0.17	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.1 8	N ₂ O-N emission factor from spring/winter-spring rice in the South*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	0.15	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.1 9	N ₂ O-N emission	N ₂ O	Emissions from agricultural by-	0.20	% kg N ₂ O-N/kg N fertilizer	Tier 2

	factor from summer-autumn/rice crop in the South*		products, livestock manure, pasture/grazing wastes			
3.20	N ₂ O-N emission factor from autumn-winter rice crop in the South*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	0.17	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.21	N ₂ O-N ³ emission factor from corn land*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	0.98	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.22	N ₂ O-N ⁴ emission factor from cassava land*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	1.52	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.23	N ₂ O-N ⁵ emission factor from sugarcane land*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	1.40	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.24	N ₂ O-N ⁶ emission factor from coffee land*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	1.76	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.25	N ₂ O-N ⁷ emission factor from	N ₂ O	Emissions from agricultural by-products, livestock manure,	2.02	% kg N ₂ O-N/kg N fertilizer	Tier 2

	rubber plantation*		pasture/grazing wastes			
3.2.6	N ₂ O-N ⁸ emission factor from tea plantation*	N ₂ O	Emissions from agricultural by-products, livestock manure, pasture/grazing wastes	1.71	% kg N ₂ O-N/kg N fertilizer	Tier 2
3.2.7	N ₂ O emission factor for forest land (EF _{2F})	N ₂ O	Direct emission of N ₂ O from organic soil	8	Kg N ₂ O-N/ha	Tier 1
3.2.8	N ₂ O emission factor for cropland and grassland (EF _{2CG})	N ₂ O	Direct emission of N ₂ O from organic soil	16	Kg N ₂ O-N/ (kg N)	Tier 1
3.2.9	N ₂ O emission factor due to evaporation (EF ₄)	N ₂ O	Indirect emissions from land/from waste management	0.01	Kg N ₂ O-N/(kg NH ₃ -N + NO _x -N due to evaporation)	Tier 1
3.3.0	N ₂ O emission factor due to leaching and leakage (EF ₅)	N ₂ O	Indirect emissions from land/from waste management	0.0075	Kg N ₂ O-N/(kg N washed out and leaked)	Tier 1
3.3.1	Scale factor to calculate the difference in water management regime before cultivation (SF _p)	CH ₄	Rice cultivation	1.00	-	Tier 1

3.3.2	Ratio factor of water regime during cultivation period (SFw) for upland rice	CH ₄	Rice cultivation	0	-	Tier 1
3.3.3	Ratio factor of water regime during cultivation period (SFw) for regularly flooded rice	CH ₄	Rice cultivation	1	-	Tier 1
3.3.4	Ratio factor of water regime during cultivation period (SFw) for intermittently flooded rice (dried once)	CH ₄	Rice cultivation	0.71	-	Tier 1
3.3.5	Ratio factor of water regime during cultivation period (SFw) for intermittently flooded rice (repeatedly dried)	CH ₄	Rice cultivation	0.55	-	Tier 1
3.3.6	Ratio factor of water regime during	CH ₄	Rice cultivation	0.54	-	Tier 1

	cultivation period (SFw) for rice irrigated by rain					
3.37	Conversion factor of organic amendment (CFOA) for straw mixed shortly (<30 days) before planting	CH ₄	Rice cultivation	1.0	-	Tier 1
3.38	Conversion factor of organic amendment (CFOA) of straw mixed for a long time (>30 days) before planting	CH ₄	Rice cultivation	0.19	-	Tier 1
3.39	Rate of organic amendment by dry weight (ROA) of spring/winter-spring rice in the North*	CH ₄	Rice cultivation	0.76	Ton/ha	Tier 2
3.40	Rate of organic amendment by dry weight (ROA) of summer-autumn/rice	CH ₄	Rice cultivation	3.72	Ton/ha	Tier 2

	crop in the North*					
3.4.1	Rate of organic amendment by dry weight (ROA) of spring/winter-spring rice in the Central region*	CH ₄	Rice cultivation	0.00	Ton/ha	Tier 2
3.4.2	Rate of organic amendment by dry weight (ROA) of summer-autumn/rice crop in the Central region*	CH ₄	Rice cultivation	0.35	Ton/ha	Tier 2
3.4.3	Rate of organic amendment by dry weight (ROA) of spring/winter-spring rice in the South*	CH ₄	Rice cultivation	0.43	Ton/ha	Tier 2
3.4.4	Rate of organic amendment by dry weight (ROA) of summer-autumn/rice	CH ₄	Rice cultivation	0.43	Ton/ha	Tier 2

	crop in the South*					
3.4 5	Rate of organic amendment by dry weight (ROA) of autumn-winter rice crop in the South*	CH ₄	Rice cultivation	0.45	Ton/ha	Tier 2
3.4 6	Rate of organic amendment by dry weight (ROA) of spring/winter-spring rice in the North*	CH ₄	Rice cultivation	0.57	Ton/ha	Tier 2
3.4 7	CH ₄ emission factor of spring/winter-spring rice in the North*	CH ₄	Rice cultivation	1.61	Kg CH ₄ /ha/day	Tier 2
3.4 8	CH ₄ emission factor of summer-autumn/rice crop in the North*	CH ₄	Rice cultivation	3.43	Kg CH ₄ /ha/day	Tier 2
3.4 9	CH ₄ emission factor of spring/winter-spring rice	CH ₄	Rice cultivation	1.92	Kg CH ₄ /ha/day	Tier 2

	in the Central region*					
3.50	CH ₄ emission factor of summer-autumn/rice crop in the Central region*	CH ₄	Rice cultivation	1.91	Kg CH ₄ /ha/day	Tier 2
3.51	CH ₄ emission factor of spring/winter-spring rice in the South*	CH ₄	Rice cultivation	1.95	Kg CH ₄ /ha/day	Tier 2
3.52	CH ₄ emission factor of summer-autumn/rice crop in the South*	CH ₄	Rice cultivation	1.83	Kg CH ₄ /ha/day	Tier 2
3.53	CH ₄ emission factor of autumn-winter rice crop in the South*	CH ₄	Rice cultivation	2.20	Kg CH ₄ /ha/day	Tier 2

(*): Country-specific greenhouse gas emission factor.

APPENDIX IV

LIST OF EMISSION FACTORS FOR GREENHOUSE GAS INVENTORY - WASTE

*(Attached to Decision No. 2626/QĐ-BTNMT dated October 10, 2022, of the
Minister of Natural Resources and Environment)*

No.	Name of greenhouse gas emission factor	Type of greenhouse gas emission	Emission source	Value	Unit	Method applied under IPCC Guidelines
1	Solid waste landfill					
1.1	Methane correction factor (MCF) of unmanaged - depth landfill (depth of 5m or more)	CH ₄	Emissions from solid waste landfill	0.80	-	Tier 2
1.2	Methane correction factor (MCF) of unmanaged - shallow landfill (depth of less than 5m)	CH ₄	Emissions from solid waste landfill	0.40	-	Tier 2
1.3	Methane correction factor (MCF) of managed - anaerobic landfill	CH ₄	Emissions from solid waste landfill	1.00	-	Tier 2
1.4	Methane correction factor (MCF) of well-	CH ₄	Emissions from solid waste landfill	0.50	-	Tier 2

	managed - semi-aerobic landfill					
1.5	Methane correction factor (MCF) of poorly managed - semi-aerobic landfill	CH ₄	Emissions from solid waste landfill	0.7	-	Tier 2
1.6	Methane correction factor (MCF) of well-managed - continuous aeration landfill	CH ₄	Emissions from solid waste landfill	0.4	-	Tier 2
1.7	Methane correction factor (MCF) of poorly managed - continuous aeration landfill	CH ₄	Emissions from solid waste landfill	0.7	-	Tier 2
1.8	Methane correction factor (MCF) of the remaining landfill	CH ₄	Emissions from solid waste landfill	0.6	-	Tier 2
1.9	Dissolved organic carbon (DOC) of feed, organic matter	CH ₄	Emissions from solid waste landfill	15	%	Tier 2
1.10	Dissolved organic	CH ₄	Emissions from solid	20	%	Tier 2

	carbon (DOC) of plants		waste landfill			
1.11	Dissolved organic carbon (DOC) of paper	CH ₄	Emissions from solid waste landfill	40	%	Tier 2
1.12	Dissolved organic carbon (DOC) of wood	CH ₄	Emissions from solid waste landfill	43	%	Tier 2
1.13	Dissolved organic carbon (DOC) of textiles	CH ₄	Emissions from solid waste landfill	24	%	Tier 2
1.14	Dissolved organic carbon (DOC) of diapers	CH ₄	Emissions from solid waste landfill	24	%	Tier 2
1.15	Dissolved organic carbon (DOC) of sludge	CH ₄	Emissions from solid waste landfill	5	%	Tier 2
1.16	Dissolved organic carbon (DOC) of industrial waste	CH ₄	Emissions from solid waste landfill	15	%	Tier 2
1.17	The fraction of DOC decomposing under anaerobic conditions (DOCf) - Non-biodegradable waste: wood, wood	CH ₄	Emissions from solid waste landfill	10	%	Tier 2

	products, twigs, etc.					
1.18	The fraction of DOC decomposing under anaerobic conditions (DOCf) - Medium degradable waste: paper, textile products, diapers.	CH ₄	Emissions from solid waste landfill	50	%	Tier 2
1.19	The fraction of DOC decomposing under anaerobic conditions (DOCf) - Biodegradable waste: leftovers, grass (garden waste except twigs)	CH ₄	Emissions from solid waste landfill	70	%	Tier 2
1.20	The percentage of CH ₄ in the gas from the landfill	CH ₄	Emissions from solid waste landfill	50	%	Tier 2
2	Biological treatment of solid waste					
2.1	Waste treatment by the method of composting - CH ₄ emission	CH ₄	Biological treatment of solid waste	10.00	g CH ₄ /kg dry waste treated	Tier 1

	factor by dry weight					
2.2	Waste treatment by the method of composting - CH ₄ emission factor by wet weight	CH ₄	Biological treatment of solid waste	4.00	gCH ₄ /kg wet waste treated	Tier 1
2.3	Waste treatment by the method of composting - N ₂ O emission factor by dry weight	N ₂ O	Biological treatment of solid waste	0.60	gN ₂ O/kg dry waste treated	Tier 1
2.4	Waste treatment by the method of composting - N ₂ O emission factor by wet weight	N ₂ O	Biological treatment of solid waste	0.24	gN ₂ O/kg wet waste treated	Tier 1
2.5	Waste treatment by anaerobic digestion at biogas facilities - CH ₄ emission factor by dry weight	CH ₄	Biological treatment of solid waste	2.00	gCH ₄ /kg dry waste treated	Tier 1
2.6	Waste treatment by anaerobic digestion at biogas facilities - CH ₄ emission	CH ₄	Biological treatment of solid waste	0.80	gCH ₄ /kg wet waste treated	Tier 1

	factor by wet weight					
2.7	Waste treatment by anaerobic digestion at biogas facilities - N ₂ O emission factor by dry weight	N ₂ O	Biological treatment of solid waste	Insignificant assumption	gN ₂ O/kg dry waste treated	Tier 1
2.8	Waste treatment by anaerobic digestion at biogas facilities - N ₂ O emission factor by wet weight	N ₂ O	Biological treatment of solid waste	Insignificant assumption	gN ₂ O/kg wet waste treated	Tier 1
3 Incineration and open-burning of waste						
3.1	Dry content according to solid waste composition of plastic	CO ₂	Waste incineration (domestic solid waste)	100	%	Tier 1
3.2	Dry content according to solid waste composition of textiles	CO ₂	Waste incineration (domestic solid waste)	80	%	Tier 1
3.3	Dry content according to solid waste composition of wood	CO ₂	Waste incineration (domestic solid waste)	85	%	Tier 1
3.4	Dry content according to solid waste	CO ₂	Waste incineration (domestic	40	%	Tier 1

	composition of diapers		solid waste)			
3.5	Dry content according to solid waste composition of food, organic matter	CO ₂	Waste incineration (domestic solid waste)	40	%	Tier 1
3.6	Dry content according to solid waste composition of plants	CO ₂	Waste incineration (domestic solid waste)	40	%	Tier 1
3.7	Dry content according to solid waste composition of paper	CO ₂	Waste incineration (domestic solid waste)	90	%	Tier 1
3.8	Total carbon fraction of plastics (CF)	CO ₂	Waste incineration (domestic solid waste)	75	%	Tier 1
3.9	Total carbon fraction of textiles (CF)	CO ₂	Waste incineration (domestic solid waste)	50	%	Tier 1
3.10	Total carbon fraction of wood (CF)	CO ₂	Waste incineration (domestic solid waste)	50	%	Tier 1
3.11	Total carbon fraction of diaper (CF)	CO ₂	Waste incineration (domestic solid waste)	70	%	Tier 1

3.12	Total carbon fraction of food, organic matter (CF)	CO ₂	Waste incineration (domestic solid waste)	38	%	Tier 1
3.13	Total carbon fraction of plants (CF)	CO ₂	Waste incineration (domestic solid waste)	49	%	Tier 1
3.14	Total carbon fraction of paper (CF)	CO ₂	Waste incineration (domestic solid waste)	46	%	Tier 1
3.15	Fossil carbon fraction in total carbon of plastics (FCF)	CO ₂	Waste incineration (domestic solid waste)	100	%	Tier 1
3.16	Fossil carbon fraction in total carbon of textiles (FCF)	CO ₂	Waste incineration (domestic solid waste)	20	%	Tier 1
3.17	Fossil carbon fraction in total carbon of diapers (FCF)	CO ₂	Waste incineration (domestic solid waste)	10	%	Tier 1
3.18	Fossil carbon fraction in total carbon of plants (FCF)	CO ₂	Waste incineration (domestic solid waste)	0	%	Tier 1
3.19	Fossil carbon fraction in total carbon	CO ₂	Waste incineration (domestic	1	%	Tier 1

	of paper (FCF)		solid waste)			
3.20	Oxidation factor OF	CO ₂	Waste incineratio n	100	%	Tier 1
3.21	CH ₄ emission factor	CH ₄	Waste incineratio n - Continuous dynamic blister incinerator	0.2	Kg CH ₄ /Gg wet waste	Tier 1
			Waste incineratio n - Continuous fluidized bed incinerator	0		
			Waste incineratio n - Semi- continuous dynamic blister incinerator	6		
			Waste incineratio n - Semi- continuous fluidized bed incinerator	188		
			Waste incineratio n - Semi- continuous dynamic blister incinerator	60		

			- mass incineration			
			Waste incineration - Fluidized bed incinerator - mass incineration	237		
3.22	N ₂ O emission factor	N ₂ O	Waste incineration - Continuous dynamic blister incinerator	47	gN ₂ O/t wet waste	Tier 1
			Waste incineration - Continuous fluidized bed incinerator	67		
			Waste incineration - Semi-continuous dynamic blister incinerator	41		
			Waste incineration - Semi-continuous fluidized bed incinerator	68		

			Waste incineration - Semi-continuous dynamic blister incinerator - mass incineration	56		
			Waste incineration - Fluidized bed incinerator - mass incineration	221		
3.23	Oxidation factor OF	CO ₂	Open-burning of waste	71	%	Tier 1
3.24	CH ₄ emission factor	CH ₄	Open-burning of waste	6,500	g/t waste by wet weight	Tier 1
3.25	N ₂ O emission factor	N ₂ O	Open-burning of waste	150	gN ₂ O/t	Tier 1
4 Wastewater treatment and discharge						
4.1	Maximum CH ₄ producing capacity - B0	CH ₄	Domestic wastewater treatment and discharge	0.60	Kg CH ₄ /kgBOD	Tier 1
4.2	Methane correction factor of a centralized, aerobic wastewater	CH ₄	Domestic wastewater treatment and discharge	0.30	-	Tier 1

	treatment system					
4.3	Methane correction factor of septic system	CH ₄	Domestic wastewater treatment and discharge	0.50	-	Tier 1
4.4	Methane correction factor of untreated domestic wastewater discharged into rivers, lakes and sea	CH ₄	Domestic wastewater treatment and discharge	0.10	-	Tier 1
4.5	N ₂ O emission factor	N ₂ O	Domestic wastewater treatment and discharge	0.01	Kg N ₂ O-N/kg-N	Tier 1
4.6	Maximum CH ₄ producing capacity - B0	CH ₄	Industrial wastewater treatment and discharge	0.25	Kg CH ₄ /kgCO ₂ D	Tier 1
4.7	Methane correction factor of a centralized, aerobic wastewater treatment system	CH ₄	Industrial wastewater treatment and discharge	0.30	-	Tier 1
4.8	Methane correction factor for semi-aerobic (shallow anaerobic)	CH ₄	Industrial wastewater treatment and discharge	0.2	-	Tier 1

	treatment system					
4.9	Methane correction factor for deep anaerobic system	CH ₄	Industrial wastewater treatment and discharge	0.8	-	Tier 1
4.10	Methane correction factor of untreated domestic wastewater discharged into rivers, lakes and sea	CH ₄	Industrial wastewater treatment and discharge	0.10	-	Tier 1

 1 Including energy consumption in transportation; energy consumption in the construction industry; energy consumption in agriculture, forestry and fisheries.

2 During clinker production, limestone is heated to produce CaO and CO₂

3 Amount of nitrogen introduced into corn land

4 Amount of nitrogen introduced into cassava land

5 Amount of nitrogen introduced into sugarcane land

6 Amount of nitrogen introduced into coffee land

7 Amount of nitrogen introduced into rubber plantation

8 Amount of nitrogen introduced into tea plantation